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| **1.** | Create a dataset in WEKA named “Student_Performance.arff”, where the dataset must have 10 attributes of StudentID (Numeric), StudyHours (Numeric), Attendance (Numeric), AssignmentScore (Numeric), MidtermMarks (Numeric), InternetAccess (Nominal- Yes/No), ExtraActivities (Nominal-Yes/No), FamilySupport (Nominal-High/Low), StressLevel (Nominal-High/Medium/Low), and Result (Class- Pass/Fail).<br><br>The created dataset must contain:<br><br>i. Missing values<br>ii. Noisy values (e.g., Attendance >100)<br>iii. Inconsistent records (e.g., negative marks)<br>iv. Duplicate rows<br><br>So that it must be usable for further Preprocessing, Classification or Association mining. | |
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| **2.** | Perform data preprocessing:<br><br>Using the dataset:<br><br>i. Identify missing values.<br>ii. Apply appropriate filter to handle missing value (s).<br>iii. Detect noisy data and normalize numeric attributes.<br>iv. Remove duplicate records.<br>v. Save cleaned dataset. | |
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| **3.** | Perform Classification task(s):<br><br>i. Apply J48 Decision Tree.<br>ii. Apply Naive Bayes classifier.<br>iii. Perform 10-fold cross-validation.<br>iv. Compare accuracy and interpret confusion matrix. | |